Task 01:

Steps

A standard ACL looks only at the source, so it goes near the destination. That means R2 Fa0/1, outbound toward the Faculty LAN.

Deny the single host, then permit everything else.

R2(config)# access-list 10 deny host 192.168.20.10

R2(config)# access-list 10 permit any

R2(config)# interface fastethernet 0/1

R2(config-f)# ip access-group 10 out

Justification: A standard ACL cannot see the destination. On R1 Fa0/1 it would block Student-PC1 from everything (Servers, Admin, Faculty), not just the Faculty LAN. Near the destination (R2 Fa0/1 out), only traffic leaving toward the Faculty LAN is filtered

Task 02:

Admin LAN 192.168.10.0/24 has the wildcard mask 0.0.0.255.

Apply the ACL to the VTY lines with access-class.

R2(config)# access-list 20 permit 192.168.10.0 0.0.0.255

R2(config)# line vty 0 4

R2(config-line)# access-class 20 in

R2(config-line)# exit

(b) The command on VTY lines is access-class. ip access-group is used on physical or logical interfaces. VTY lines are virtual terminal lines, not interfaces, so there is no interface to apply it to. access-class filters the source of incoming Telnet/SSH sessions to the router itself.

(c) Verification:

R2# show access-lists 20

The output shows (n matches) next to each line.

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Task 03:

Create the standard named ACL, add the remark, deny the Student LAN, permit any.

Place it near the destination: R2 Fa0/0 (toward the servers), outbound.

R2(config)# ip access-list standard STUDENTS_NO_SERVERS

R2(config-std-nacl)# remark Block Student LAN from Server Farm

R2(config-std-nacl)# deny 192.168.20.0 0.0.0.255

R2(config-std-nacl)# permit any

R2(config-std-nacl)# exit

R2(config)# interface fastethernet 0/0

R2(config-f)# ip access-group STUDENTS_NO_SERVERS out

Justification: Standard ACL, so near the destination. An outbound ACL on Fa0/0 filters only traffic leaving toward the Server Farm. Admin and Faculty are permitted by permit any.

(c) Yes, Students can still ping 172.16.1.1. That packet is destined to R2 itself and is never forwarded out Fa0/0, so the outbound ACL does not apply to it.

Task 04:

(a) Working

Range size = 127 − 64 + 1 = 64 addresses, which is a block of 64, so /26 (255.255.255.192).

network address = 192.168.20.64

Wildcard = 255.255.255.255 − 255.255.255.192 = 0.0.0.63

Twist: Only one ACL per direction per interface is allowed. If you apply FACULTY_PROTECT on R2 Fa0/1 out, it replaces ACL 10. Student PC1 (.10) is not in the .64–.127 range, so it would be unblocked and Task 01 would be lost. The fix is one combined ACL.

(b) Combined ACL

R2(config)# ip access-list standard FACULTY_PROTECT

R2(config-std-nacl)# remark Block Student-PCl and Student range .64-.127 from Faculty
R2(config-std-nacl)# deny host 192.168.2.0.10
R2(config-std-nacl)# deny 192.168.20.64 0.0.0.63
R2(config-std-nacl)# permit any
(c) Swap on the interface
R2(config)# interface fastethernet 0/1
R2(config-f)# no ip access-group 10 out
R2(config-f)# ip access-group FACULTY_PROTECT out

Task 05:
Extended ACL, so near the source: R1 Fa0/1, inbound (traffic entering from the Student LAN).
Deny TCP 20 and 21 to the FTP server, then permit all other IP.
R1(config)# access-list 110 deny tcp 192.168.20.0 0.0.0.255 host 172.16.1.20 eq 20
R1(config)# access-list 110 deny tcp 192.168.20.0 0.0.0.255 host 172.16.1.20 eq 21
R1(config)# access-list 110 permit ip any any
R1(config)# interface fastethernet 0/1
R1(config-if)# ip access-group 110 in
Justification: An extended ACL near the source drops unwanted packets before they cross the WAN. Inbound on Fa0/1 filters them as soon as they enter R1.
(c) No. Port 21 is the control connection (login and commands). If it is blocked, the session cannot be established, so no file transfer can start. Port 20 is only used for the data channel in active mode.
Task 06:
Near the source: R2 Fa0/1, inbound.
Order: the specific permit for Faculty PC2 first, then the broader SMTP deny, then Telnet deny, then permit any.

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R2(config)# access-list 120 permit tcp host 192.168.30.11 host 172.16.1.30 eq 25
R2(config)# access-list 120 deny tcp 192.168.30.0 0.0.0.255 host 172.16.1.30 eq 25
R2(config)# access-list 120 deny tcp 192.168.30.0 0.0.0.255 any eq 23
R2(config)# access-list 120 permit ip any any
R2(config)# interface fastethernet 0/1
R2(config-f)# ip access-group 120 in

(c) If the SMTP deny line comes first, Faculty-PC2's packets also match the broad deny (the whole /24 includes .11). The permit line is never reached, so Faculty-PC2 is blocked too. ACLs are matched top-down and stop at the first match.

Task 07:

Deny ICMP echo (type 8) from the Student LAN to the Admin LAN, then permit everything else.

Apply near the source: R1 Fa0/1, inbound.

R1(config)# access-list 130 deny icmp 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255 echo
R1(config)# access-list 130 permit ip any any
R1(config)# interface fastethernet 0/1
R1(config-f)# ip access-group 130 in

(b) Yes, Admin-PC1 can ping Student-PC2.

The echo request enters R1 on Fa0/0 and leaves out Fa0/1. The ACL is inbound on Fa0/1, so the request never meets it. Student-PC2 sends the echo-reply (ICMP type 0), which enters R1 on Fa0/1 and is checked by ACL 130. The deny line matches only echo (type 8), so the reply does not match it. It falls to permit ip any any and is delivered.

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Task 08

R1(config)# ip access-list extended ADMIN_FILTER

R1(config-ext-nacl)# deny tcp host 192.168.10.11 host 172.16.1.20 eq ftp

R1(config-ext-nacl)# deny tcp host 192.168.10.11 host 172.16.1.20 eq ftp-data

R1(config-ext-nacl)# deny tcp 192.168.10.0 0.0.0.255 host 172.16.1.30 eq smtp

R1(config-ext-nacl)# permit ip any any

R1(config-ext-nacl)# exit

R1(config)# interface fastethernet 0/0

R1(config-if)# ip access-group ADMIN_FILTER in

Justification: Extended ACL, so near the source. The traffic comes from the Admin LAN, so it is dropped at R1 Fa0/0 inbound.

Task 09

(a)

R2(config)# ip access-list extended NO_ECHO_REPLY

R2(config-ext-nacl)# deny icmp 172.16.1.0 0.0.0.255 192.168.20.0 0.0.0.255 echo-reply

R2(config-ext-nacl)# permit ip any any

R2(config-ext-nacl)# exit

R2(config)# interface fastethernet 0/0

R2(config-if)# ip access-group NO_ECHO_REPLY in

(b) Yes, the echo request reaches the Web Server. The server sends a reply, but R2 drops it. The student sees "Request timed out".

(c) Task 07 generates less unnecessary traffic. It drops the request at the source, so nothing crosses the WAN or reaches the server. In Task 09 the request travels the whole path, the server processes it and replies, and only then is the reply dropped.

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Task IV

(a)

R2(config)# ip access-list extended FACULTY_WEB_ONLY

R2(config-ext-nacl)# permit tcp 192.168.30.0 0.0.0.255 host 172.16.1.10 eq www

R2(config-ext-nacl)# permit tcp 192.168.30.0 0.0.0.255 host 172.16.1.10 eq 443

R2(config-ext-nacl)# permit udp 192.168.30.0 0.0.0.255 host 172.16.1.40 eq domain

R2(config-ext-nacl)# deny ip 192.168.30.0 0.0.0.255 172.16.1.0 0.0.0.255

R2(config-ext-nacl)# permit ip any any

R2(config-ext-nacl)# exit

R2(config)# interface fastethernet 0/0

R2(config-f)# ip access-group FACULTY_WEB_ONLY out

(b) R2 Fa0/1 inbound already has ACL 120 (Task 06). only one ACL is allowed per direction per interface, so this would replace it.

(c) Packets from Faculty travel to R2 and are routed first, then dropped at the last step. This wastes router processing compared with dropping them on entry.

Task V

(a)

R1(config)# ip access-list extended MGMT_ACCESS

R1(config-ext-nacl)# permit tcp host 192.168.10.10 any eq 22

R1(config-ext-nacl)# deny tcp any any eq telnet log

R1(config-ext-nacl)# exit

R1(config)# line vty 0 4

R1(config-line)# access-class MGMT_ACCESS in

R1(config-line)# transport input ssh

(b) Log sends a syslog message every time a packet matches the

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TASK 1C

(a)

R2(config)# ip access-list extended FACULTY_WEB_ONLY

R2(config-ext-nacl)# 35 permit tcp host 192.168.30.10 host 172.16.1.20 eq ftp

R2(config-ext-nacl)# 36 permit tcp host 192.168.30.10 host 172.16.1.20 eq ftp-data

R2(config-ext-nacl)# no 20

R2(config-ext-nacl)# exit

R2(config)# interface fastethernet 0/0

R2(config-f)# no ip access-group FACULTY_WEB_ONLY out

R2(config-f)# exit

R2(config)# no ip access-list extended FACULTY_WEB_ONLY

R2(config)# end

R2# show access-lists FACULTY_WEB_ONLY

R2# show ip interface fastethernet 0/0

R2# show running-config | include access

(show access-lists = contents with hit counts, show ip interface = applied ACL, show running-config | include access = ACL lines in running config.)

(b) Sequence 40 denies all Faculty traffic to the Server Farm. ACLs are checked top-down, so FTP lines placed after 40 would never be reached.

Task 3

(a)

Permit / Deny --> Matched Line

Deny --> Line 1 (the /24 includes .12)

Deny --> Line 3

Permit --> Line 4

Deny --> Line 5

Deny --> Implicit deny

Deny --> Implicit deny

(b)

R1(config)# no access-list 150

R1(config)# access-list 150 permit tcp host 192.168.20.12 host 172.16.1.20 eq 21

R1(config)# access-list 150 permit tcp host 192.168.20.12 host 172.16.1.20 eq 20

R1(config)# access-list 150 deny tcp 192.168.20.0 0.0.0.255 host 172.16.1.20 eq 21

R1(config)# access-list 150 deny icmp 192.168.20.0 0.0.0.127 any echo

R1(config)# access-list 150 deny udp any any eq 53

R1(config)# access-list 150 permit ip any any

The PC3 permit must come before the deny, and the final permit ip any any allows everything not explicitly denied.

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1. Yes. The same ACL can be applied to several interfaces, each with its own ip access-group command. For example, STUDENTS_NO_SERVERS could go on both R2 Fa0/0 out and R2 Fa0/1 out.

2. Implicit deny drops anything not explicitly permitted, so the ACL is secure by default. An ACL with only deny statements blocks all traffic.

3. Yes. One ACL per protocol, per direction, per interface, so at most one inbound and one outbound IPv4 ACL on an interface. A second one in the same direction replaces the first.

4. Standard ACLs check only the source, so near the source they would block that source from everything. They go near the destination. Extended ACLs match exact traffic, so near the source they drop unwanted packets early and save bandwidth.

5. Yes. ACLs are matched top-down and stop at the first match. If permit any is first, everything is permitted and the later lines never match.

6. A subnet mask uses 1s for the network part. A wildcard mask uses 0 for "must match" and 1 for "ignore", so it is the inverse. /26 = 0.0.0.63, /28 = 0.0.0.15, /20 = 0.0.15.255, host = 0.0.0.0.

7. No. A standard ACL filters only the source IP, so it cannot tell FTP from web traffic. An extended ACL is needed.

8. ip access-group is applied on interfaces and filters traffic passing through the router. access-class is applied on VTY lines and filters Telnet/SSH access to the router itself.

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9. No. outbound ACLs do not filter packets generated by the router itself, only transit traffic.

10. Blocking echo stops ping requests, so the hosts that send pings cannot ping the destination. Blocking echo-reply lets requests arrive but drops the answers. The hosts being pinged cannot reply, and the pinger sees timeouts.

11. no access-list 110 deny tcp ... removes the whole ACL 110. To remove one line safely, use ip access-list extended 110, check sequence numbers with show access-lists, then no <sequence>.

12. Named ACLs: (1) descriptive names, (2) individual lines can be inserted or deleted using sequence numbers, (3) easier to edit and read, with remarks.

13. No. Traffic between PCs on the same switch is switched at Layer 2 and never reaches the router interface.

14. Inbound is more efficient. Denied packets are dropped before the routing lookup, while outbound ACLs are applied after routing.

15. remark documents the purpose of the ACL and does not affect filtering. log generates a syslog message when a line matches, for monitoring.

16. The ACL is treated as empty, so all traffic is permitted.

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